

THOMAS REESE

Full-Stack Software Engineer · AI Systems & Developer Tooling

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SUMMARY

Full-stack software engineer (4+ yrs) building production AI systems, agentic **LLM** tooling, and developer infrastructure. Currently at Envoy, where I built **Codebase Intelligence**: an 87,500-chunk vector index serving 22 repos through an **MCP** server on **AWS Bedrock**. Production stack spans Ruby on Rails, Ember.js, React 19, and Python — comfortable across **Pulumi/Terraform** IaC, **RAG** retrieval, **LLM evals**, and tool-use guardrails.

PROFESSIONAL EXPERIENCE

Envoy | Full-Stack Software Engineer, Visitor Management San Francisco, CA | Nov 2024 – Present

- Built **Codebase Intelligence**, a **RAG** platform indexing 22 internal repos (~87,500 AST-aware code chunks, tree-sitter parsed) on **Amazon S3 Vectors** and **Bedrock Knowledge Bases (Titan Text Embeddings V2)**, with a Python **MCP server** exposing six retrieval tools to Anthropic **Claude**. Hit **Hit@1 = 0.785** on a 200-query human-labeled eval set at ~\$0.03/month, replacing per-engineer 6–8 GB local vector DBs.
- Migrated the visitor-management dashboard from **Ember/Glimmer** to **React 19** across six surfaces (profile, printers, screeners, visits, matching-rules, documents) using **Module Federation 2.0**, **TanStack Query v5**, and **Radix UI**, shipping incrementally behind feature flags.
- Designed **AgentLock**, a phase-based **guardrails** and permission model for the 14 **agentic** AI tools embedded in the Visitors product — capability gating per request lifecycle, observability hooks, and policy enforcement at the MCP boundary.
- Delivered **Visitor Profile Merge** and **Block List V2** end-to-end across two quarters: Rails APIs, React merge UI, and the identity-matching lifecycle (expiration, disable toggle, search, deletion). Validated with TDD: **RSpec** on Rails, **Vitest + MSW** on React, plus staging-pod-exec mutation testing.
- Replaced fragile SKU-name string matching with a **ChargeBee feature-code subscription gate** on the Visitors Basic plan: counter table, gate-state API, 75% / 100% threshold mailer, and reset cron, spanning Rails and Ember in one sprint.
- Authored Envoy's internal AI playbook: AI-tooling onboarding guide, a six-layer agent memory reference, and a coding-agent **observability** writeup adopted by Envoy's Engineering Innovation Program.
- Sustained **5+ PRs/week** across nine repos on a stacked-PR (**Graphite**) and worktree workflow, shipping Rails, Ember (Octane / Glimmer / GTS), React 19, **Cerbos** authorization, and Pulumi / Terraform IAM.

Sprout Social | Front-End Engineer

Chicago, IL | May 2022 – Apr 2024

- Migrated core front-end components from **Flow** to **TypeScript**, eliminating a class of runtime type errors and unblocking strict-mode adoption across the codebase.
- Built and maintained the **Seeds** shared React component library, standardizing UI primitives across product teams and cutting per-feature build time by ~3 days.
- Delivered a React UI for **bulk sentiment reclassification** (1,000 messages per batch) using **Redux** and virtualized lists, cutting analyst handle time ~30% per batch.
- Built a **Query Builder** in React + TypeScript backed by **OpenAI GPT** keyword suggestions with debounced search, lowering query-setup time for social-listening analysts.
- Launched a **faceted message-exploration tool** (React Context, Redux, React Router) with virtualized lists, lifting session engagement ~30% on the listening dashboard.

The Walt Disney Company | Software Engineering Intern

Remote | May 2021 – Nov 2021

- Built **ESPN's Banner-Maker**: reusable React components with Redux state and team / player API integration, cutting editorial banner-creation time ~35%.
- Owned the React front-end for a **Student Loan Refinance** product: responsive UI, client-side validation, and back-end API integration, cutting form-validation errors ~60%.
- Drove **WCAG** accessibility work with **Applito**: screen-reader audits and semantic-HTML refactors that raised automated a11y coverage ~70%.

Credit Karma | Full-Stack Web Developer Intern

San Francisco, CA | Jan 2020 – Dec 2020

- Built internal full-stack tooling across **React**, **Node.js**, **Ruby**, and **Python** services on **AWS**, focused on infrastructure automation and front-end developer experience.

SKILLS

Languages:	TypeScript, JavaScript (ES6+), Python, Ruby, HTML5, CSS3, SQL, Flow (legacy)
Frameworks & UI:	React 19, Next.js, Ember.js (Octane / Glimmer / GTS), Ruby on Rails, FastAPI, Node.js, Express, Tailwind, Radix UI, SCSS, Luxon, Three.js, Framer Motion
AI & LLM:	Model Context Protocol (MCP), Anthropic Claude SDK, OpenAI SDK, Gemini Flash, Amazon Bedrock, S3 Vectors, Titan Text Embeddings V2, RAG, agentic tool use, LLM evals, prompt engineering, function calling, AI guardrails, tree-sitter AST chunking, LangChain, LangGraph
Vector / Data:	ChromaDB, Pinecone, pgvector, PostgreSQL, Redis, Kafka, Sidekiq, MongoDB, Optuna
Cloud & Infra:	AWS (Bedrock, S3, IAM), Pulumi, Terraform, Kubernetes, Docker, OpenTelemetry, Datadog, Vercel
Build & State:	Module Federation 2.0, Vite, Webpack 5, TanStack Query v5, Redux Toolkit, Axios, MSW v2, GraphQL, REST
Testing:	Vitest, RSpec, Playwright, Jest, Cypress, MSW v2, mutation testing
Workflow & Sec:	Graphite (stacked PRs), GitHub Actions, CircleCI, Cerbos, OAuth, JWT, WCAG accessibility

EDUCATION

Dominican University	River Forest, IL
B.S. Computer Science	Aug 2018 – May 2022

PROJECTS

TraderDan — Automated Trading System [github]	Python · Next.js · FastAPI · Anthropic Claude · TradeLocker · PostgreSQL · Optuna · ChromaDB
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Live multi-asset trading system across five active prop-firm accounts (XAUUSD H1/M15 via TradeLocker). Architecture: backtest engine, FastAPI webhook execution server with prop-firm risk gates (daily-loss, drawdown, ATR-spike, news blackout, weekend closer), a 14-agent AI team led by an OpenClaw orchestrator (Anthropic Claude tool use, Gemini Flash routing), an eval harness with drift detection, ChromaDB semantic memory, and a Next.js 15 dashboard. 5,000+ tests, stacked-PR workflow.

Blue Ridge Canine Academy [link]	React · Node.js · Express · MongoDB · Acuity Scheduling · Figma
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Freelance build (\$800) for a local dog trainer: a full-stack scheduling and payments platform. React frontend, Node.js / Express backend, MongoDB persistence. Wireframed in Figma and hosted on GoDaddy.

HONORS & AWARDS

- **1st place**, Google × Code Nation Hackathon.
- **Won 24-hour Code for Good Hackathon** (JP Morgan), leading a front-end team shipping a site for the Girls in the Game non-profit.
- **2nd place**, ACCA Programming Contest — highest Dominican University finish to date. Completed **MLT Career Prep**; STEM mentor at **Code Nation**.